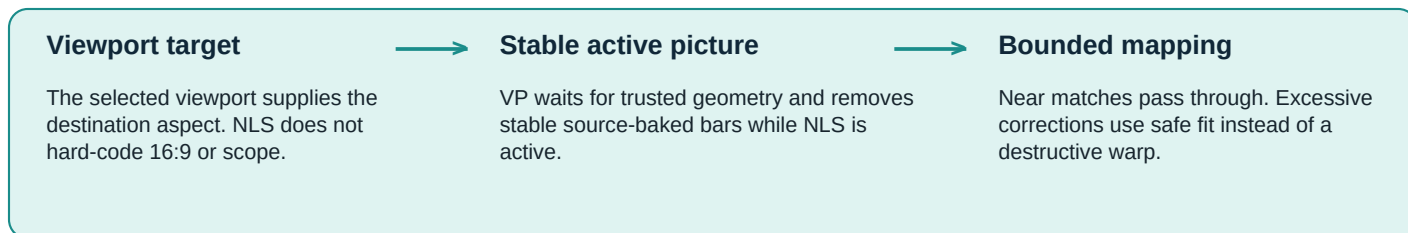


VideoProcessor NLS

Configuration Guide

Standard NLS, NLS+, and NLS-V - supported behavior and safe tuning

What NLS does



Choose a mode

Standard NLS

One selected axis

VP + DIRECTSHOW

Best compatibility choice for 4:3 to 16:9 or 16:9 to a scope viewport. Classic is the supplied baseline; protected geometry can reserve more of the center.

NLS+

Balanced two-axis sharing

VP RENDERER ONLY

Shares a bounded amount across both axes. This reduces side concentration but adds center zoom and vertical motion; the zoom cap limits aggressive requests.

NLS-V

Vertical one-axis mapping

VP RENDERER ONLY

For content wider than target, typically scope to 16:9. It reuses NLS.gsl vertically and can apply symmetric side crop to hold warp below the chosen limit.

The key tradeoff

NLS cannot create a distortion-free fill when source and target aspects differ. Standard NLS concentrates correction toward one pair of edges; NLS+ distributes some of it across both axes; presentation crop discards a bounded amount of real picture to reduce the correction. Choose which compromise is least visible for your material.

Configure safely

Config application workflow

- Open Shader Groups and use the NLS group. The root selection is the reversible Off state.
- Create or edit a member with `shader_type` set to `nls`. Keep modes exclusive unless you intentionally build a multi-effect group.
- Set the VP Renderer shader file. For NLS+ and NLS-V, leave the DirectShow shader file blank because those modes are unsupported there.
- Add mapping values in Custom parameters. Blank backend file fields are valid and mean that backend ignores the member.
- Assign a unique shortcut or an automatic when rule. Save, activate the mode, and verify the selected member in diagnostics.

Target and bar handling

- Use the normal selected viewport. Do not create an NLS-specific viewport or renderer.
- `screen_aspect` is the physical destination target, not a content label. NLS compares each stable active picture with that target.
- When VP Renderer NLS is active, trusted encoded bars are removed from the NLS source even if the viewport's ordinary `automatic_crop` setting is off.
- Optional presentation crop is additional real-picture crop. It applies only while an eligible VP Renderer NLS mapping is active.
- NLS Off preserves ordinary viewport presentation. Waiting, direction bypass, invalid geometry, and safe-fit fallback do not apply intentional crop.

Supported profile examples

STANDARD NLS - current baseline

```
[shader.nls.standard]
shader_type: nls
label: Nonlinear Stretch
stage: post_resize
hlsl_file: NLS.hlsl
glsl_file: NLS.glsl
quality: high
geometry: classic
strength: 1.0
curve: 2.0
tolerance_percent: 5
max_stretch_ratio: 1.4
aspect_direction: narrower_only
vprenderer_max_crop_percent: 0
```

NLS+ - current VP Renderer defaults

```
[shader.nls.plus]
shader_type: nls
label: NLS+
stage: pre_resize
glsl_file: NLSPlus.glsl
geometry: protected
strength: 1.0
horizontal_center_protection: 0.35
vertical_center_protection: 0.25
curve: 2.0
axis_balance: 0.25
max_center_zoom: 1.08
tolerance_percent: 2
max_stretch_ratio: 1.4
aspect_direction: any
vprenderer_max_crop_percent: 2
vprenderer_crop_preference: minimize_distortion
```

NLS-V - current VP Renderer defaults

```
[shader.nls.vertical]  shader_type: nls  label: NLS-V  stage: pre_resize
glsl_file: NLS.glsl  geometry: protected  strength: 1.0  center_protection: 0.35
curve: 2.0  tolerance_percent: 5  max_stretch_ratio: 1.22  aspect_direction: wider_only
vprenderer_max_crop_percent: 6  vprenderer_crop_preference: preserve_image
```

Common NLS parameters

Parameter	Valid values	Meaning and tuning effect
geometry	classic protected	Classic uses a smooth one-axis falloff. Protected reserves a central radius, then concentrates transition nearer the edges.
strength	0 to 1	Standard NLS: overall nonlinear intensity. NLS+: requested complementary-axis sharing only; total aspect correction remains complete.
center_protection	0 to 0.45	Standard protected NLS and NLS-V only. Higher values protect more center area but force a stronger transition outside it.
curve	1.0 to 2.9	2.0 is the supplied baseline. Higher values concentrate correction nearer the edges and can make side motion more obvious.
quality	high	Bundled high-quality shader mode. Separate from the VP Renderer global quality setting.
tolerance_percent	0 to 50; usually 2 to 5	Aspect suitability guard, not strength. Near matches pass through. Too high can engage NLS where the mismatch is visually negligible.
max_stretch_ratio	1.0 to 1.5	Hard policy limit. Above it VP uses complete-source safe fit rather than partially warping an unsafe correction.
aspect_direction	narrower_only wider_only any	Eligibility guard. narrower_only maps toward a wider target; wider_only supports NLS-V; any derives the axis from measured geometry.
vprenderer_max_crop_percent	0 to 10 per affected edge	VP Renderer only. Zero preserves all picture pixels. Nonzero allows symmetric real-picture crop to reduce the correction.
vprenderer_crop_preference	preserve_image minimize_distortion	preserve_image uses only enough crop to meet the stretch cap. minimize_distortion uses the greatest valid configured crop without crossing the target aspect.

NLS+ only

Parameter	Valid / supplied	Meaning
axis_balance	0 to 1 / 0.25	Requests how much correction moves to the complementary axis. 0 is one-axis behavior; 0.5 requests an equal logarithmic split; the zoom cap can reduce the effective value.
max_center_zoom	1.0 to 1.25 / 1.08	Caps the uniform center enlargement caused by two-axis sharing. 1.0 collapses NLS+ to one-axis correction; values above 1.08 permit more sharing and more visible zoom.
horizontal_center_protection	0 to 0.45 / 0.35	Protected horizontal radius. Larger values keep center behavior over more width and move the transition toward the sides.
vertical_center_protection	0 to 0.45 / 0.25	Protected vertical radius. The smaller supplied value spreads the vertical transition to reduce top/bottom breathing during pans.

Recommended tuning order

1

Confirm the viewport target

If screen_aspect is wrong, every later NLS decision is wrong.

2

Set direction and stretch limit

Use narrower_only, wider_only, or any; then choose the largest correction you consider acceptable.

3

Choose geometry and protection

Start with supplied classic or protected values. Do not tune several shape controls at once.

4

Tune NLS+ sharing

Start at axis_balance 0.25 and max_center_zoom 1.08. Raise only after watching faces, circles, pans, and titles.

5

Add crop last

Crop trades picture content for less distortion. Start at 0, then use the smallest amount that materially helps.

6

Verify transitions

Test aspect changes, menus, subtitles, cuts, and channel changes - not only a paused test pattern.

Limits and drawbacks

- No NLS mode is geometry-neutral. Straight lines, circles, faces, camera pans, and edge tickers can reveal the warp.
- NLS+ reduces edge concentration by adding complementary-axis scaling. Its characteristic artifacts are center zoom and top/bottom breathing; it may simply be less pleasing for some content.
- Center protection does not remove distortion. It moves more of the transition outside the protected region, which can make edges accelerate more strongly.
- Presentation crop removes real image pixels. minimize_distortion may use the entire allowed crop; preserve_image uses only what is needed to meet the stretch cap.
- If permitted crop cannot produce a safe ratio, VP discards the tentative crop and uses complete-source safe fit. It never applies an over-limit partial warp.
- NLS depends on trusted, stable active-picture geometry. During waiting or unstable transitions, safe presentation is expected and temporary non-engagement is not a shader failure.
- DirectShow renderers require shader support and use standard one-axis NLS only. NLS+, NLS-V selection, and intentional presentation crop are VP Renderer-only.

Troubleshooting cues

Symptom	Try first
Side motion or edge squeeze	Lower max_stretch_ratio, compare NLS+ at axis_balance 0.25, or allow a very small VP Renderer crop.
Center zoom is too obvious	Lower max_center_zoom or axis_balance. A value of 1.0 for max_center_zoom reduces NLS+ to one-axis behavior.
Top/bottom breathing	Lower NLS+ strength or axis_balance; keep vertical_center_protection near the supplied 0.25 baseline.
Picture edges are missing	Reduce vprenderer_max_crop_percent or use preserve_image instead of minimize_distortion.
NLS does not engage	Check the selected viewport target, aspect_direction, tolerance, trusted geometry, and whether the requested ratio exceeds the stretch limit.

Practical default: If NLS+ does not look better to you, use Standard NLS. NLS+ is an optional tradeoff, not an upgrade in every scene.